

Dark-Field Bi-Telecentric Inspection System: Full-Field Defect Detection in Circular Sapphire Watch Glass Components

Project Overview

This project focuses on the design of an optical inspection system for defect detection in circular sapphire glass samples used as protective cover components in wearable consumer electronics. The sample has a circular geometry with a diameter of approximately 4 cm, requiring a large field of view capable of inspecting the entire component in a single acquisition.



Figure 1. Sample Layout, Sapphire glass for wearable consumer electronics.

To address this requirement, a bi-telecentric lens was selected to provide a large and stable field of view while minimizing perspective distortion. The proposed system uses dark-field illumination to enhance defect visibility in transparent sapphire samples, where defects such as scratches, cracks, inclusions, edge defects, or surface contamination may appear as bright scattering features against a dark background.

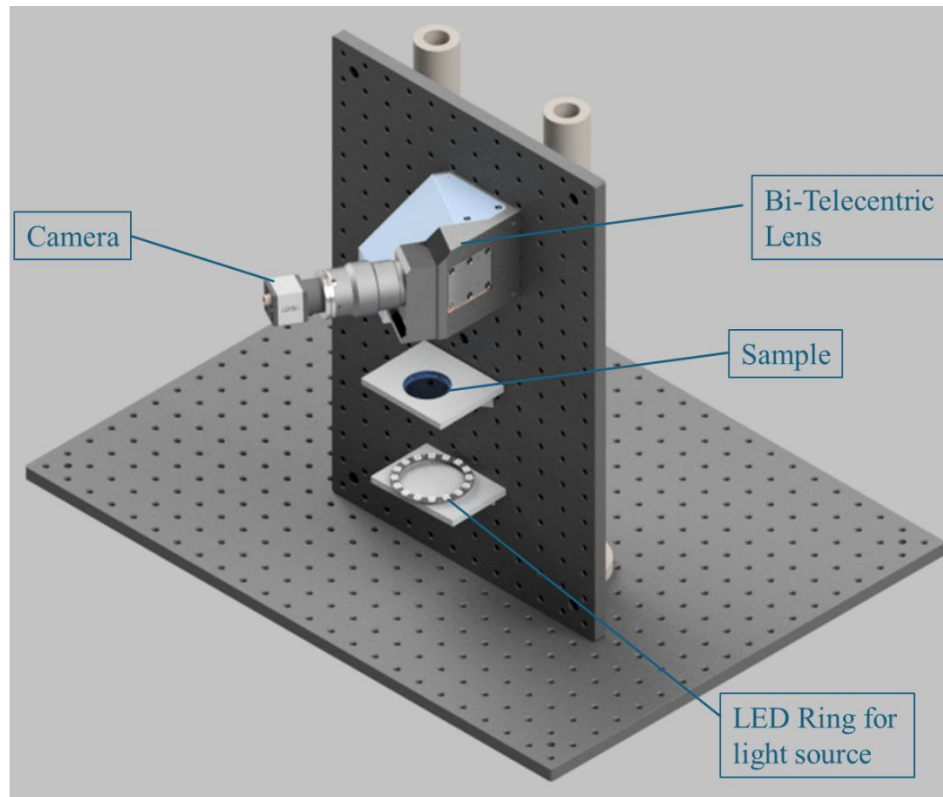


Figure 2. System Layout.

Motivation

The main challenge of this project is that the entire sapphire sample must be inspected within a single field of view. Since sapphire is an anisotropic material, polarization-based imaging produced a full-field optical response from the sample itself, making it difficult to isolate localized defects.

A standard bright-field imaging setup may not provide enough contrast for transparent or weakly scattering defects, while higher-magnification optical systems may not cover the full 4 cm circular sample in one image.

The inspection system must provide:

- Large field of view for a 4 cm circular sample
- Low distortion across the image
- High contrast for localized defect detection
- Compatibility with transparent sapphire glass
- Potential for future automated in-line defect detection

Initial Testing

The first approach used crossed linear polarizers to block the transmitted light and make defects appear as brighter regions. However, the sapphire sample itself introduced polarization changes across the full field, likely due to its material properties. As a result, the entire sample produced a polarization response, making localized defects difficult to isolate.

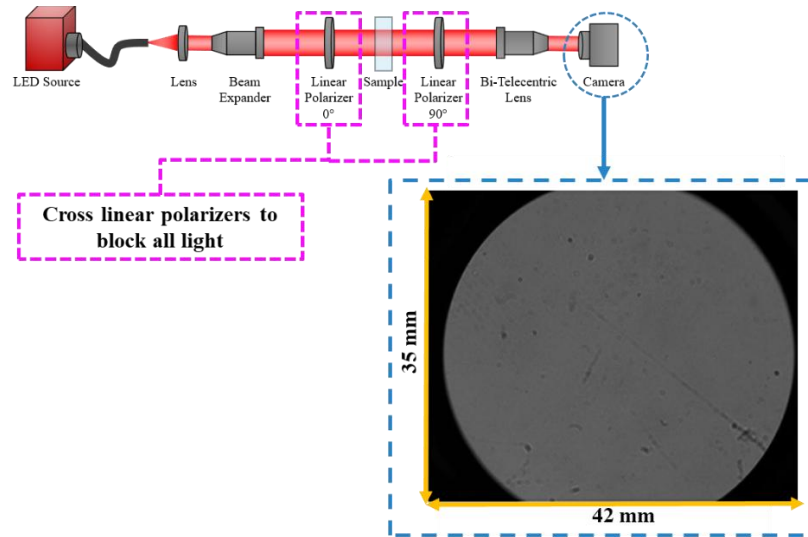


Figure 3. Initial test and preliminary results.

A photoelasticity-based measurement was also tested using polarization imaging to estimate phase retardance. This provided useful information about the optical anisotropy of the sample, but it was not sufficient for reliable defect detection because the material response appeared across the whole sample.

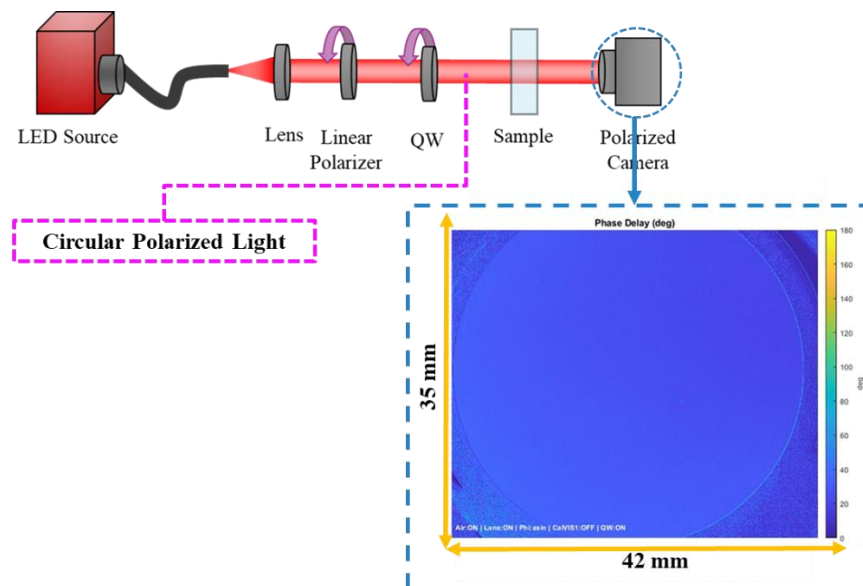


Figure 4. Initial test and preliminary results.

Proposed Solution

Based on these results, a dark-field imaging approach was selected. In dark-field inspection, the direct transmitted light is blocked or minimized, while light scattered by defects is collected by the imaging system. This allows defects to appear as bright features on a dark background.

The combination of dark-field illumination and a bi-telecentric lens allowed defect visibility while maintaining full-field inspection of the sapphire sample.

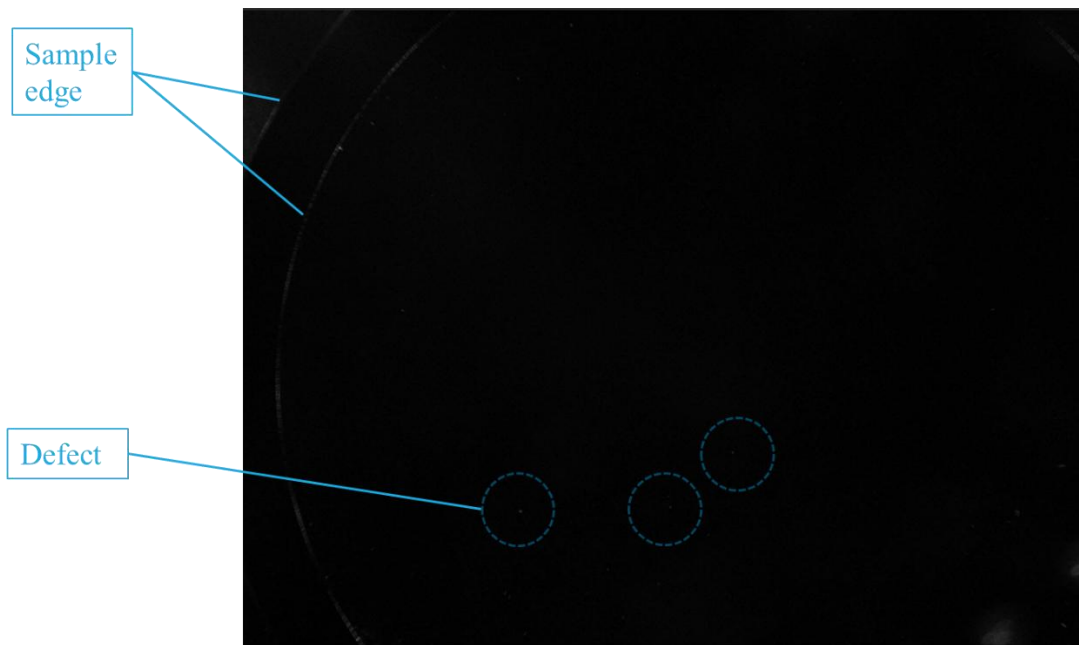


Figure 5. Dark field test and preliminary results.

My Role

My contributions to this project included:

- Designed the full-field optical inspection concept.
- Evaluated crossed-polarizer and photoelasticity-based approaches.
- Identified the limitation of polarization-based inspection for this sapphire sample.
- Proposed dark-field imaging as a more suitable defect detection method.
- Selected a bi-telecentric lens configuration for large-field inspection.
- Prepared technical presentations to communicate the experimental results and next steps.

Key Technical Areas

- Dark-field imaging
- Bi-telecentric lens inspection
- Transparent material inspection
- Sapphire glass defect detection
- Polarization imaging
- Photoelasticity-based preliminary testing
- Optical system design
- Large-field-of-view imaging

Current Status

Preliminary imaging confirmed that the bi-telecentric lens can provide a field of view large enough to inspect the circular sapphire sample. Crossed-polarizer and phase retardance measurements showed that the sample produces a full-field polarization response, making localized defect detection difficult with polarization-based methods alone.

The project is currently moving toward a dark-field imaging configuration to improve defect contrast. Future work will focus on developing custom software for automated defect detection and inline inspection.

Confidentiality Note

Some technical details, raw data, and partner information are omitted or generalized due to confidentiality considerations. The project is described as a full-field inspection system for circular sapphire glass components provided by an external consumer electronics partner.